

NAME

mbmesh – Process swath bathymetry data and generate 3D point clouds and GLB models

SYNOPSIS

mbmesh [-I *datalist*] [-O *outputdir*] [-R *west/east/south/north*] [-html --html] [-V [-V]] [-H]

DESCRIPTION

mbmesh reads swath sonar data files and processes bathymetric soundings to generate 3D point cloud data in multiple formats. The primary use is to extract multibeam echosounder (MBES) data and convert it to point clouds suitable for 3D visualization and analysis.

mbmesh reads a datalist file containing references to swath data files in MB-System supported formats. It extracts all valid bathymetry soundings, applies optional geographic filtering, and outputs the data in multiple formats:

adjustedPointcloud.xyz

Point cloud in local projected coordinates (meters) relative to data centroid

ecefPointcloud.xyz

Point cloud in WGS84 Earth-Centered Earth-Fixed (ECEF) coordinates

adjustedPointcloud.glb

Binary GLB (glTF) format suitable for 3D viewer applications

adjustedPointcloud.html

Optional HTML viewer for interactive 3D visualization

OPTIONS**-I *datalist***

Input datalist file containing list of swath data files. This is a required option. Each line in the datalist file should contain: path format [processed_path] following MB-System datalist format conventions. The default datalist file is *datalist.mb-1* if not specified.

-O *outputdir*

Output directory for all generated files. The directory will be created if it does not exist. Default is *./tileset*.

-R *west/east/south/north*

Specify geographic bounds for filtering soundings. Only soundings within the specified rectangle (in degrees longitude/latitude) will be included in the output. The format is west/east/south/north where west and south are typically negative. If not specified, all soundings are included.

-html

Also generate an HTML viewer file (*adjustedPointcloud.html*) for interactive 3D visualization of the point cloud. When this option is used, **mbmesh** will attempt to launch an HTTP server and open the viewer in the default web browser.

-V

Increase verbosity level. By default, only essential information is printed. Can be specified multiple times (**-V -V**) to increase debugging output. Higher verbosity levels provide detailed progress information, data statistics, and sample soundings from the dataset.

-H

Display help message and exit.

-h

Display help message and exit (alternative to **-H**).

EXAMPLES

Process all files in a datalist and output to default directory:

mbmesh -Idatalist.mb-1

Process files in a specific geographic region:

mbmesh -Idatalist.mb-1 -R-122.5/-121.8/36.5/37.2 -V

Process files with custom output directory and HTML viewer:

mbmesh -Idatalist.mb-1 -O./my_output -html -V

Process files with increased verbosity for detailed debugging:

mbmesh -Idatalist.mb-1 -V -V

OUTPUT FILES

mbmesh generates the following output files in the specified output directory:

adjustedPointcloud.xyz

Standard XYZ point cloud in local projected coordinates where longitude/latitude are converted to meters relative to the data centroid and depth is centered around the mean depth. This format is suitable for visualization in standard 3D point cloud viewers. Format: X(meters) Y(meters) Z(meters)

ecefPointcloud.xyz

Point cloud in WGS84 ECEF coordinates with GeoOrigin offset applied. This format preserves accurate Earth-centered coordinates suitable for analysis requiring geodetic precision. Format: X(meters) Y(meters) Z(meters)

adjustedPointcloud.glb

Binary GLB (GL Transmission Format) file containing the point cloud as a 3D model. This format is widely supported by web-based 3D viewers and applications. Can be directly embedded in web pages or used with tools like Cesium.js or Three.js for interactive visualization.

adjustedPointcloud.html

Optional HTML file generated when **-html** option is used. This file provides an interactive web-based viewer for the GLB model. When generated, **mbmesh** attempts to launch a local HTTP server and open the viewer in the default web browser.

DATA STATISTICS

Upon completion, **mbmesh** prints detailed statistics including:

Number of datalist files processed and successfully read

Total pings processed from all files

Total beams encountered vs. valid beams (acceptance rate)

Geographic bounds of the processed data

Sample soundings (first 10) with coordinates and quality flags

PROCESSING NOTES

Beam Quality Filtering

mbmesh automatically filters out beams flagged as bad or suspect using MB-System's standard beam quality flags. Only beams marked as good are included in the output point clouds. The percentage of valid beams vs. total beams is reported in the statistics.

Geographic Filtering

When **-R** is specified, soundings outside the geographic bounding box are excluded from output. This is useful for processing large datasets or focusing on areas of interest.

Coordinate Transformations

The adjusted point cloud uses a local projected coordinate system relative to the data centroid for better numerical stability in 3D applications. The ECEF point cloud preserves absolute WGS84 coordinates. Both formats include reference information in file headers for reproducibility.

Memory Usage

mbmesh stores all soundings in memory during processing. For very large datasets, this may require significant RAM. The program reports total soundings collected upon completion.

MB-SYSTEM INTEGRATION

mbmesh integrates with the MB-System datalist processing framework, allowing it to process multiple files in any MB-System supported format. File format is determined automatically or can be specified via the datalist file. Processed files (if available) are preferred over raw files unless the datalist indicates otherwise.

FUTURE DEVELOPMENT

Planned enhancements to **mbmesh** include:

- Triangle mesh generation from point cloud with seafloor feature preservation
- Py3dtiles integration and evaluation of alternative 3D Tiles libraries
- Draco compression support for efficient GLB file transmission
- Improved X3DOM visualization and viewer enhancements
- Refinement of GeoOrigin handling and direct ECEF coordinate output

AUTHOR

David W. Caress and Dale N. Chayes
Monterey Bay Aquarium Research Institute
Original implementation: CSUMB Capstone - Spring 2026

SEE ALSO

mb-system(1), **mbgrid(1)**, **mbgrd2gltf(1)**, **mbgrdviz(1)**